

Nasir Yasin

Norfolk, VA,United States.

+1 (757) 992-1759

Google Scholar

nasiryasin1

nyasin@odu.edu

About me

Ph.D. Candidate in Computational and Applied Mathematics at Old Dominion University, specializing in machine learning, Python-based data analysis, scientific computing, and scalable numerical modeling.

Skills

- Python, MATLAB, R, C++, Fortran
- CUDA, Linux, Numerical Modeling
- LaTeX, Microsoft 365, Tecplot;
- Visual Studio Code, Github, Git
- Jupyter Notebook

Awards

- KNU International Graduate Scholarship (2021–2023)
- BK21 Plus Scholarship (2022)
- Prime Minister's National Laptop Scheme Award (2018)

Certifications

- Deep Learning with MATLAB
- Machine Learning with MATLAB
- Python For Machine Learning
- MATLAB for MATH and Data Science
- 3D Digital Modeling

Language

- English, Pashto, Urdu, Hindi,Punjabi

Hobbies

- Soccer
- Cricket

Interests

AI & Machine Learning, Algorithm Development, Scientific Computing, Numerical Analysis and Modeling, Data Assimilation

Education

- Ph.D. in Computational and Applied Mathematics *2023–Present*
  Old Dominion University, USA
  Concentration: *Mathematics of Data Science*
- M.S. in Computational and Applied Mathematics *2023–2025*
  Old Dominion University, USA
  Thesis: *Entropy and Gradient Stable Scheme for Hyperbolic Burgers Equation*
- Master of Science in Mathematics *2019–2021*
  COMSATS University Islamabad, Pakistan
  Thesis: *Flow Features and Force Statistics Analysis of Flow Past Two Offset Cylinders.*
- B.S. in Mathematics *2014–2018*
  International Islamic University Islamabad, Pakistan

Experience

- Aug 2023–present: Graduate student in Applied & Computational Mathematics at ODU, focusing on numerical methods, HPC, and machine learning; SI Leader tutoring and undergraduate math courses.
- Apr 2022–Aug 2023 Applied Fluid Laboratory, KNU, South Korea, Research Assistant on NRF-funded project; simulated high-Re flows using LBM and implemented GPU-accelerated C++ solvers for oscillating-wing energy recovery.
- Jan 2020 - Apr 2022: (HEC-funded project) Research Assistant, COMSATS University Islamabad ; SRT-LBM numerical simulations developed in Fortran with post-processing and visualization in MATLAB and Tecplot.

Projects

- Jan 2024–Aug 2025: Entropy and Gradient Stable Scheme for Hyperbolic Burgers Equations
- Dec 2023: Convergence and Performance Comparison of Numerical Solvers for Sparse Resistor-Network Linear Systems
- Dec 2023: Image Processing and Compression Using Multiple Algorithms.
- April 2022– August 2023: Improved energy recovery performance of oscillated wings using articulated and flexible bodies
- Jan 2020–Apr 2022: Numerical study of wake and thrust around Multiple Cylinders at moderate Reynolds numbers at various configurations

Publications

- 2025 – Investigations of conjugate heat transfer and fluid flow in partitioned porous cavity using Darcy–Forchheimer model: FEM base computations
- 2024 – A novel computational analysis of boundary-driven two-dimensional heat flow with internal heat generation
- 2024 – Flow patterns behind staggered square cylinders using the Lattice Boltzmann method: numerical and statistical study

Conferences and seminars

- Investigation of Flow Dynamics Around Bluff Bodies Using SRT-LBM presented at 50th Annual New York State Regional Graduate Mathematics Conference.
- Flow Structures Around Bluff Bodies at Low Reynolds Numbers, presented at the 15th Annual Math Awareness Conference, Old Dominion University, 2025.
- Flow Patterns Behind Stationary and Moving Bluff Bodies Utilizing the SRT-Lattice Boltzmann Method presented at SEARCDE 2024.